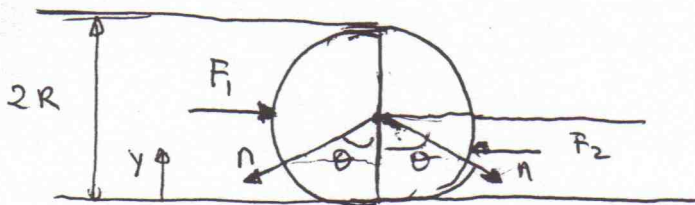
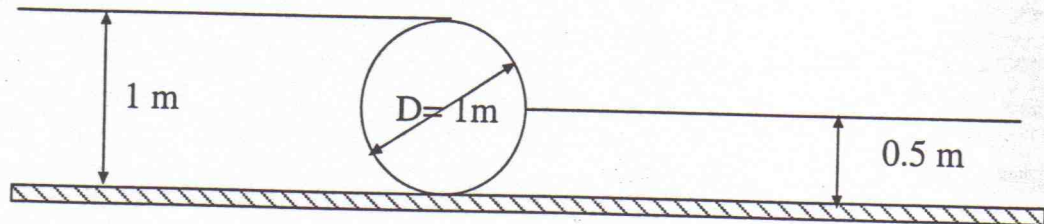


4- Calcule la fuerza neta horizontal sobre el cilindro de 5 metros de longitud que se muestra en la figura adjunta.



$$ds = LR d\theta$$

$$n_x = -\sin\theta$$

$$n_x = \sin\theta$$

$$\begin{aligned} \bar{F}_1 &= - \int (P - P_a) \bar{n}_x ds = LR \int_0^{\pi} \rho g (2R - y) \sin\theta d\theta \\ &= \int_0^{\pi} \rho g (2R - \underbrace{(R + R \cos\theta)}_y) \sin\theta d\theta = L \int_0^{\pi} R^2 \rho g (1 + \cos\theta) \sin\theta d\theta \\ &= \rho g R^2 L \int_0^{\pi} (\sin\theta + \sin\theta \cos\theta) d\theta = \rho g R^2 L (-\cos\theta + \frac{1}{2} \sin^2\theta) \Big|_0^{\pi} \end{aligned}$$

$$\boxed{F_1 = 2 \rho g R^2 L}$$

$$\begin{aligned} \bar{F}_2 &= - \rho g L R \int_0^{\pi/2} (R - y) \sin\theta d\theta = - \rho g L R^2 \int_0^{\pi/2} \cos\theta \sin\theta d\theta \\ &= - \rho g L R^2 \frac{\sin^2\theta}{2} \Big|_0^{\pi/2} = \boxed{- \frac{\rho g L R^2}{2} = F_2} \end{aligned}$$

$$F_T = F_1 - F_2 = \boxed{\frac{3}{2} \rho g L R^2 = F_T}$$